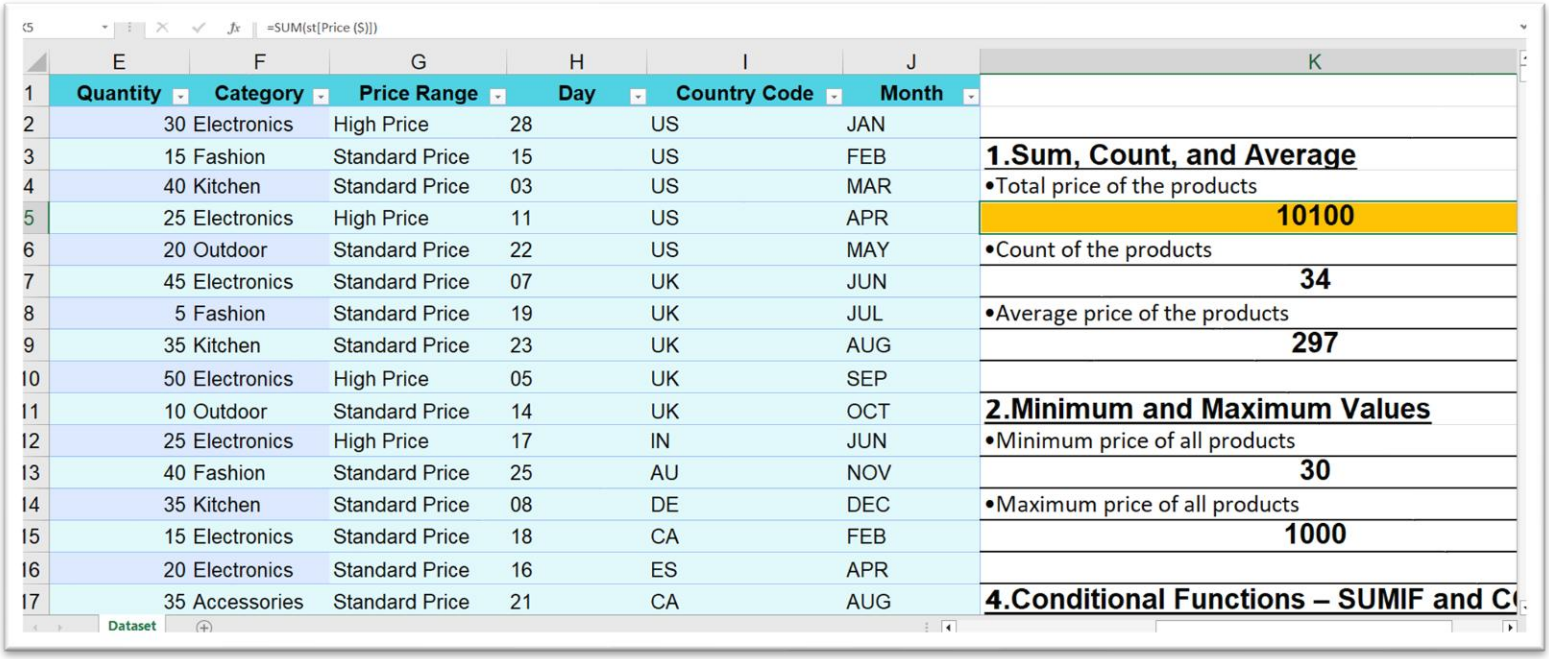
Excel Assignment 1 – (Data Exploration)

1. Sum, Count, and Average

USING SUM FORMULA

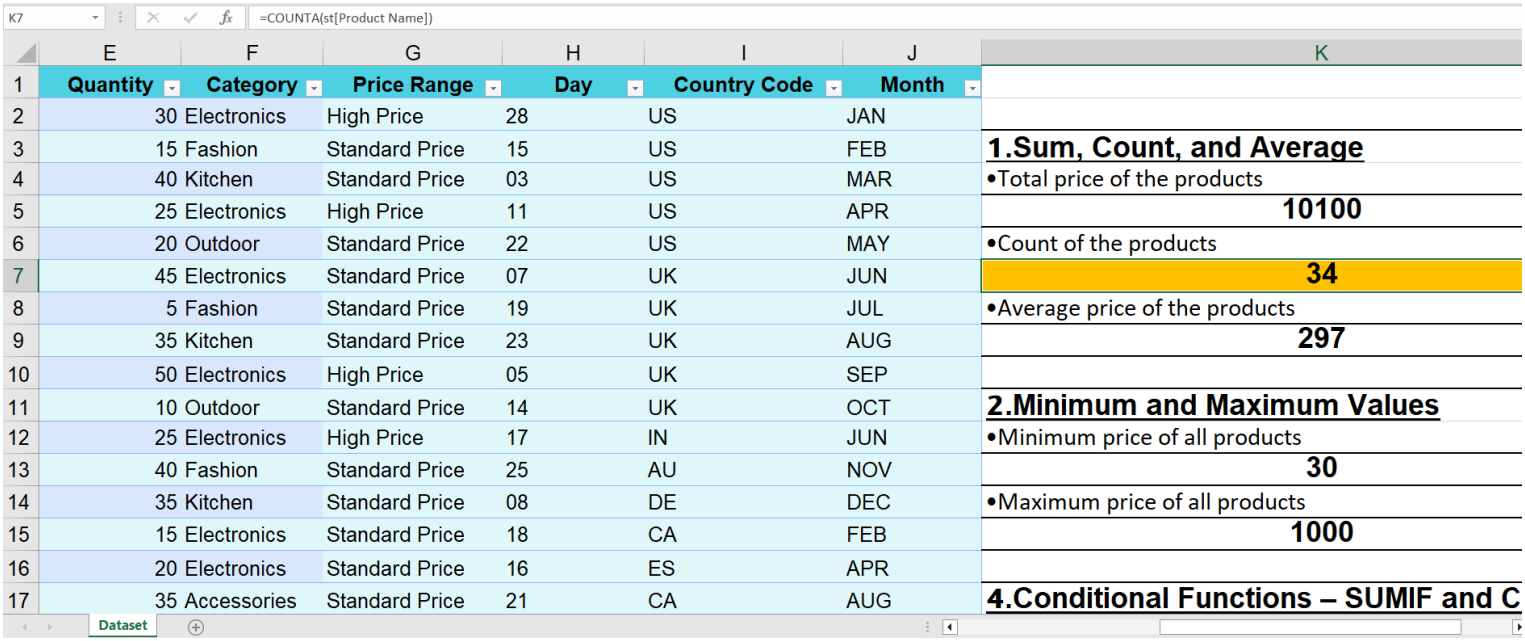


=SUM(st[Price ($)])

	E	F	G	H	I	J	K
1	Quantity	Category	Price Range	Day	Country Code	Month	
2	30	Electronics	High Price	28	US	JAN	
3	15	Fashion	Standard Price	15	US	FEB	1.Sum, Count, and Average
4	40	Kitchen	Standard Price	03	US	MAR	•Total price of the products
5	25	Electronics	High Price	11	US	APR	10100
6	20	Outdoor	Standard Price	22	US	MAY	•Count of the products
7	45	Electronics	Standard Price	07	UK	JUN	34
8	5	Fashion	Standard Price	19	UK	JUL	•Average price of the products
9	35	Kitchen	Standard Price	23	UK	AUG	297
10	50	Electronics	High Price	05	UK	SEP	
11	10	Outdoor	Standard Price	14	UK	OCT	2.Minimum and Maximum Values
12	25	Electronics	High Price	17	IN	JUN	•Minimum price of all products
13	40	Fashion	Standard Price	25	AU	NOV	30
14	35	Kitchen	Standard Price	08	DE	DEC	•Maximum price of all products
15	15	Electronics	Standard Price	18	CA	FEB	1000
16	20	Electronics	Standard Price	16	ES	APR	
17	35	Accessories	Standard Price	21	CA	AUG	4.Conditional Functions – SUMIF and C

Dataset

USING COUNTA FORMULA

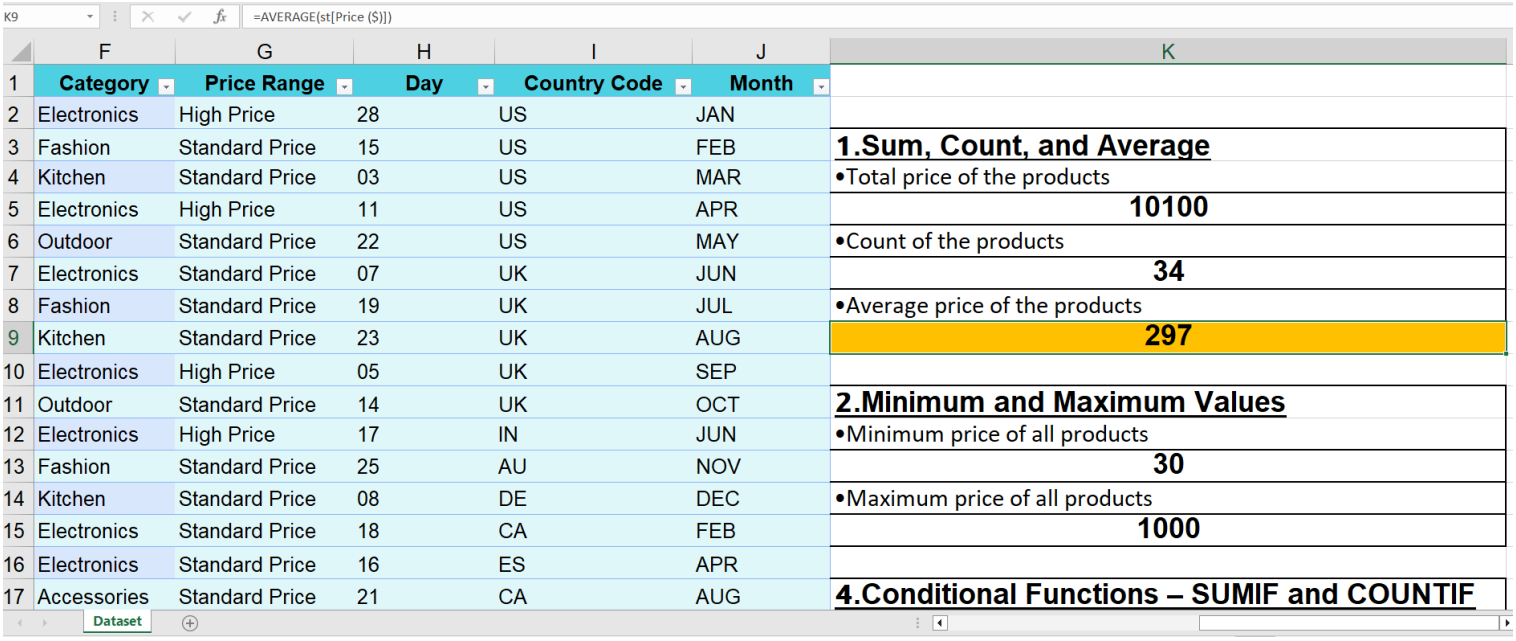


K7 =COUNTA(st[Product Name])

	E	F	G	H	I	J	K
1	Quantity	Category	Price Range	Day	Country Code	Month	
2	30	Electronics	High Price	28	US	JAN	
3	15	Fashion	Standard Price	15	US	FEB	1.Sum, Count, and Average
4	40	Kitchen	Standard Price	03	US	MAR	•Total price of the products
5	25	Electronics	High Price	11	US	APR	10100
6	20	Outdoor	Standard Price	22	US	MAY	•Count of the products
7	45	Electronics	Standard Price	07	UK	JUN	34
8	5	Fashion	Standard Price	19	UK	JUL	•Average price of the products
9	35	Kitchen	Standard Price	23	UK	AUG	297
10	50	Electronics	High Price	05	UK	SEP	
11	10	Outdoor	Standard Price	14	UK	OCT	2.Minimum and Maximum Values
12	25	Electronics	High Price	17	IN	JUN	•Minimum price of all products
13	40	Fashion	Standard Price	25	AU	NOV	30
14	35	Kitchen	Standard Price	08	DE	DEC	•Maximum price of all products
15	15	Electronics	Standard Price	18	CA	FEB	1000
16	20	Electronics	Standard Price	16	ES	APR	
17	35	Accessories	Standard Price	21	CA	AUG	4.Conditional Functions – SUMIF and C

Dataset

USING AVERAGE FORMULA

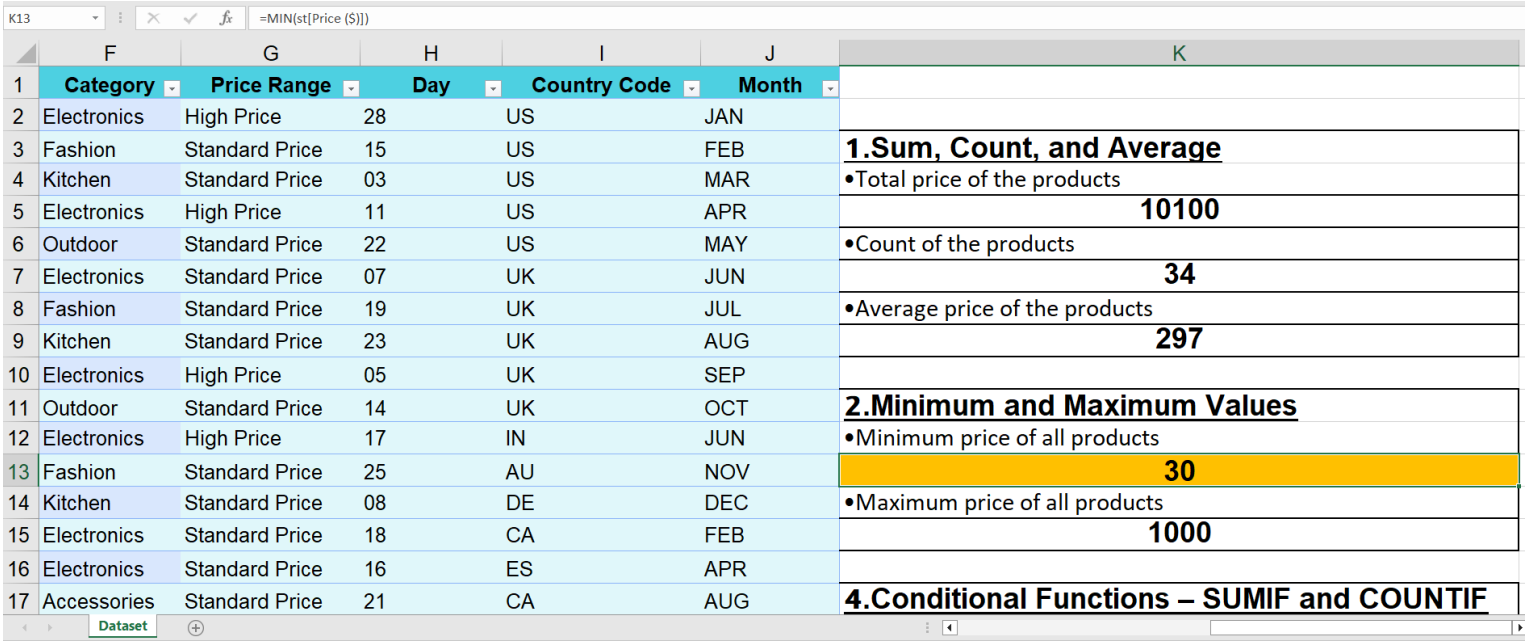


	F	G	H	I	J	K
1	Category	Price Range	Day	Country Code	Month	
2	Electronics	High Price	28	US	JAN	
3	Fashion	Standard Price	15	US	FEB	1.Sum, Count, and Average
4	Kitchen	Standard Price	03	US	MAR	•Total price of the products
5	Electronics	High Price	11	US	APR	10100
6	Outdoor	Standard Price	22	US	MAY	•Count of the products
7	Electronics	Standard Price	07	UK	JUN	34
8	Fashion	Standard Price	19	UK	JUL	•Average price of the products
9	Kitchen	Standard Price	23	UK	AUG	297
10	Electronics	High Price	05	UK	SEP	
11	Outdoor	Standard Price	14	UK	OCT	2.Minimum and Maximum Values
12	Electronics	High Price	17	IN	JUN	•Minimum price of all products
13	Fashion	Standard Price	25	AU	NOV	30
14	Kitchen	Standard Price	08	DE	DEC	•Maximum price of all products
15	Electronics	Standard Price	18	CA	FEB	1000
16	Electronics	Standard Price	16	ES	APR	
17	Accessories	Standard Price	21	CA	AUG	4.Conditional Functions – SUMIF and COUNTIF

Dataset

2.Minimum and Maximum Values

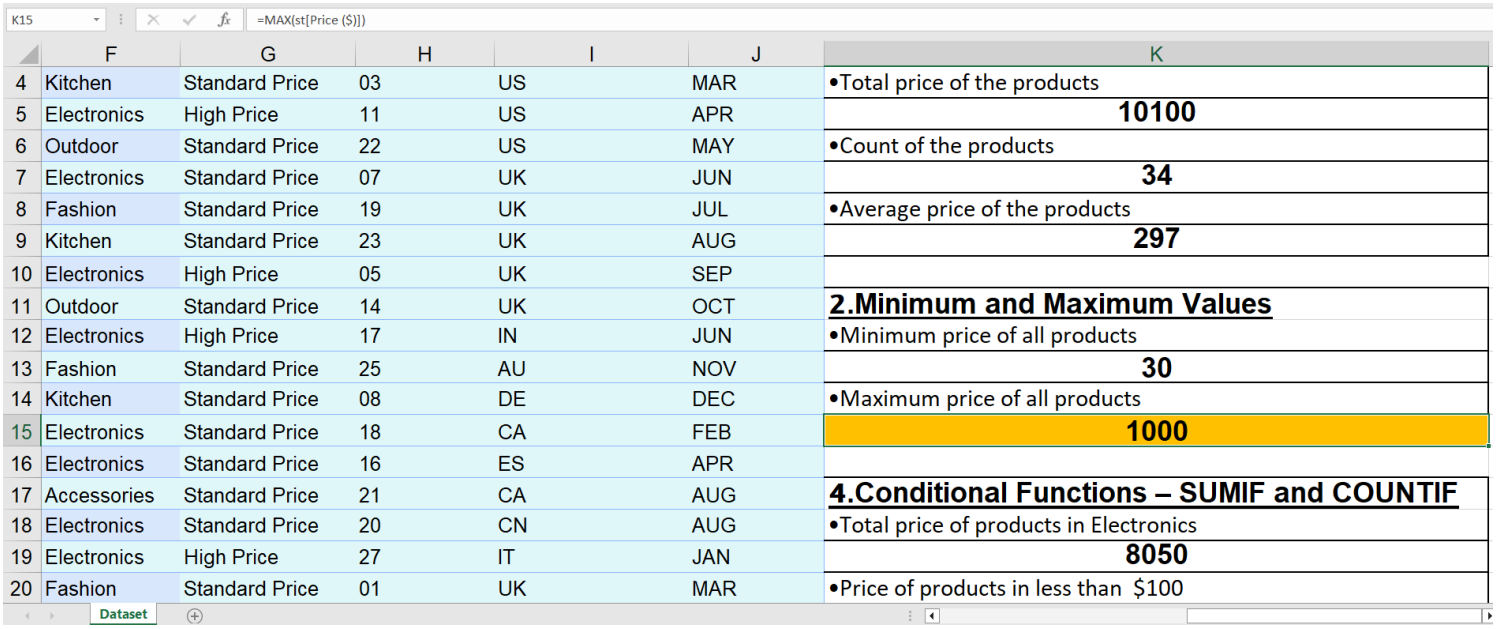
USING MINIMUM FORMULA



	F	G	H	I	J	K
1	Category	Price Range	Day	Country Code	Month	
2	Electronics	High Price	28	US	JAN	
3	Fashion	Standard Price	15	US	FEB	1.Sum, Count, and Average
4	Kitchen	Standard Price	03	US	MAR	•Total price of the products
5	Electronics	High Price	11	US	APR	10100
6	Outdoor	Standard Price	22	US	MAY	•Count of the products
7	Electronics	Standard Price	07	UK	JUN	34
8	Fashion	Standard Price	19	UK	JUL	•Average price of the products
9	Kitchen	Standard Price	23	UK	AUG	297
10	Electronics	High Price	05	UK	SEP	
11	Outdoor	Standard Price	14	UK	OCT	2.Minimum and Maximum Values
12	Electronics	High Price	17	IN	JUN	•Minimum price of all products
13	Fashion	Standard Price	25	AU	NOV	30
14	Kitchen	Standard Price	08	DE	DEC	•Maximum price of all products
15	Electronics	Standard Price	18	CA	FEB	1000
16	Electronics	Standard Price	16	ES	APR	
17	Accessories	Standard Price	21	CA	AUG	4.Conditional Functions – SUMIF and COUNTIF

Dataset

USING MAXIMUM FORMULA



K15 =MAX(st[Price ($)])

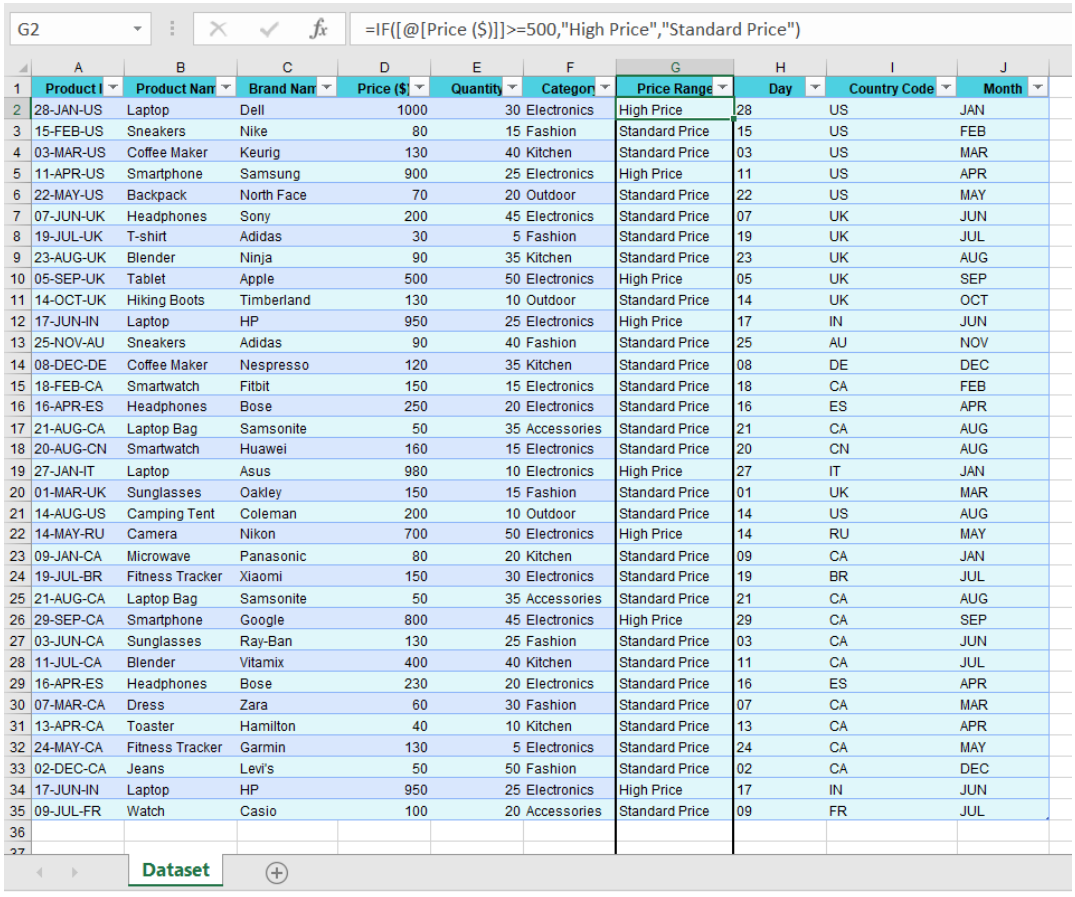
	F	G	H	I	J	K
4	Kitchen	Standard Price	03	US	MAR	•Total price of the products
5	Electronics	High Price	11	US	APR	**10100**
6	Outdoor	Standard Price	22	US	MAY	•Count of the products
7	Electronics	Standard Price	07	UK	JUN	**34**
8	Fashion	Standard Price	19	UK	JUL	•Average price of the products
9	Kitchen	Standard Price	23	UK	AUG	**297**
10	Electronics	High Price	05	UK	SEP	
11	Outdoor	Standard Price	14	UK	OCT	**2.Minimum and Maximum Values**
12	Electronics	High Price	17	IN	JUN	•Minimum price of all products
13	Fashion	Standard Price	25	AU	NOV	**30**
14	Kitchen	Standard Price	08	DE	DEC	•Maximum price of all products
15	Electronics	Standard Price	18	CA	FEB	**1000**
16	Electronics	Standard Price	16	ES	APR	
17	Accessories	Standard Price	21	CA	AUG	**4.Conditional Functions – SUMIF and COUNTIF**
18	Electronics	Standard Price	20	CN	AUG	•Total price of products in Electronics
19	Electronics	High Price	27	IT	JAN	**8050**
20	Fashion	Standard Price	01	UK	MAR	•Price of products in less than $100

Dataset

3. Logical Function – IF

USING IF FORMULA - NEW COLUMN

- If Price ≥ $500 → classify as High Price
- If Price < $500 → classify as Standard Price



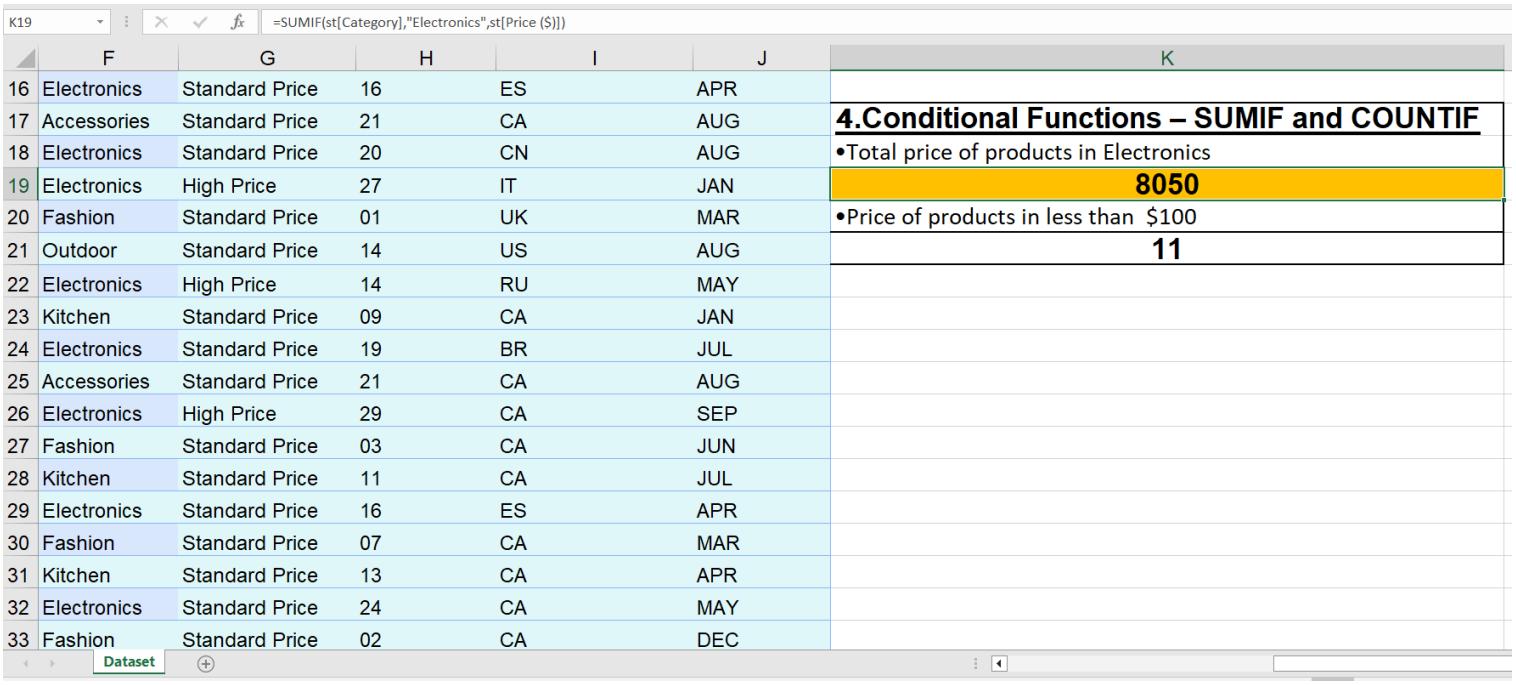
G2 =IF([@[Price ($)]]>=500,"High Price","Standard Price")

	A	B	C	D	E	F	G	H	I	J
1	Product I	Product Nan	Brand Nan	Price ($	Quantity	Categor	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
35	09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price	09	FR	JUL
36										

Dataset

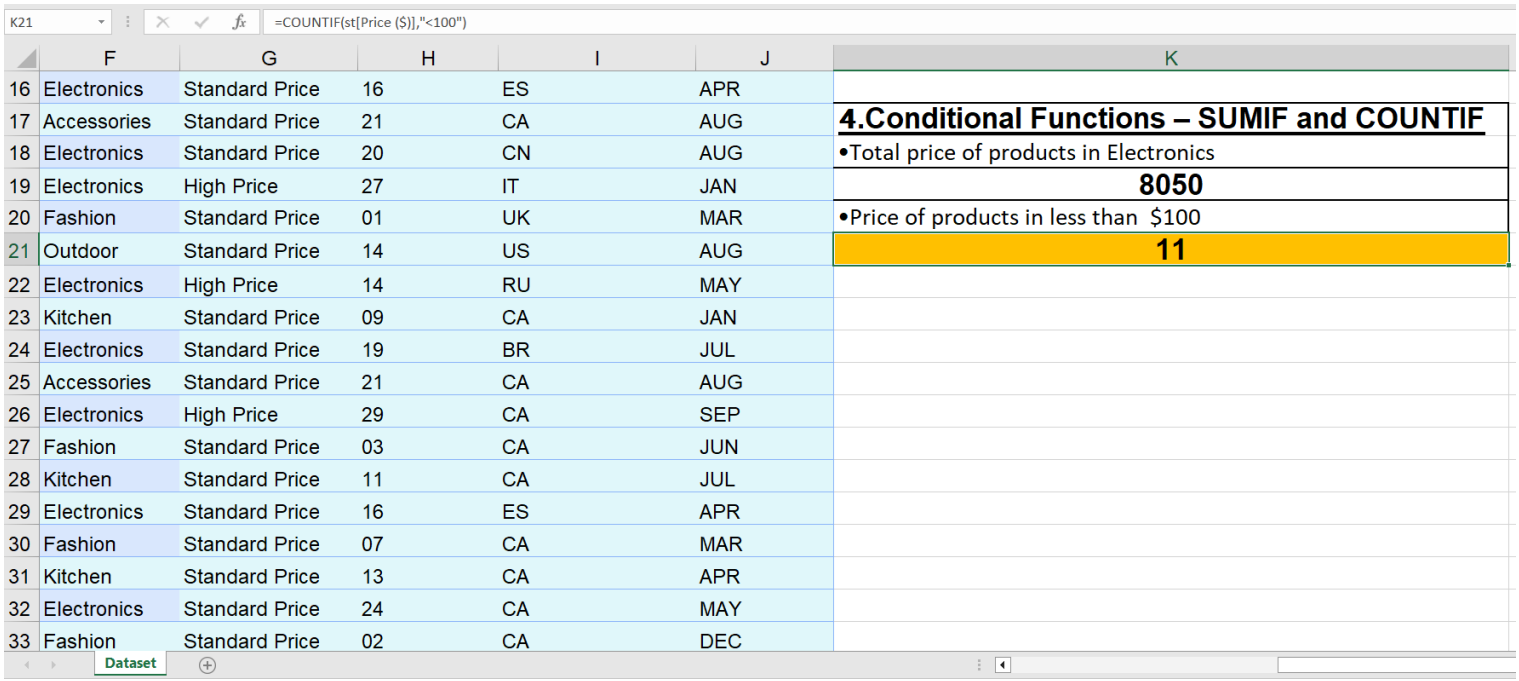
4. Conditional Functions – SUMIF and COUNTIF

USING SUMIF FORMULA



	F	G	H	I	J
16	Electronics	Standard Price	16	ES	APR
17	Accessories	Standard Price	21	CA	AUG
18	Electronics	Standard Price	20	CN	AUG
19	Electronics	High Price	27	IT	JAN
20	Fashion	Standard Price	01	UK	MAR
21	Outdoor	Standard Price	14	US	AUG
22	Electronics	High Price	14	RU	MAY
23	Kitchen	Standard Price	09	CA	JAN
24	Electronics	Standard Price	19	BR	JUL
25	Accessories	Standard Price	21	CA	AUG
26	Electronics	High Price	29	CA	SEP
27	Fashion	Standard Price	03	CA	JUN
28	Kitchen	Standard Price	11	CA	JUL
29	Electronics	Standard Price	16	ES	APR
30	Fashion	Standard Price	07	CA	MAR
31	Kitchen	Standard Price	13	CA	APR
32	Electronics	Standard Price	24	CA	MAY
33	Fashion	Standard Price	02	CA	DEC

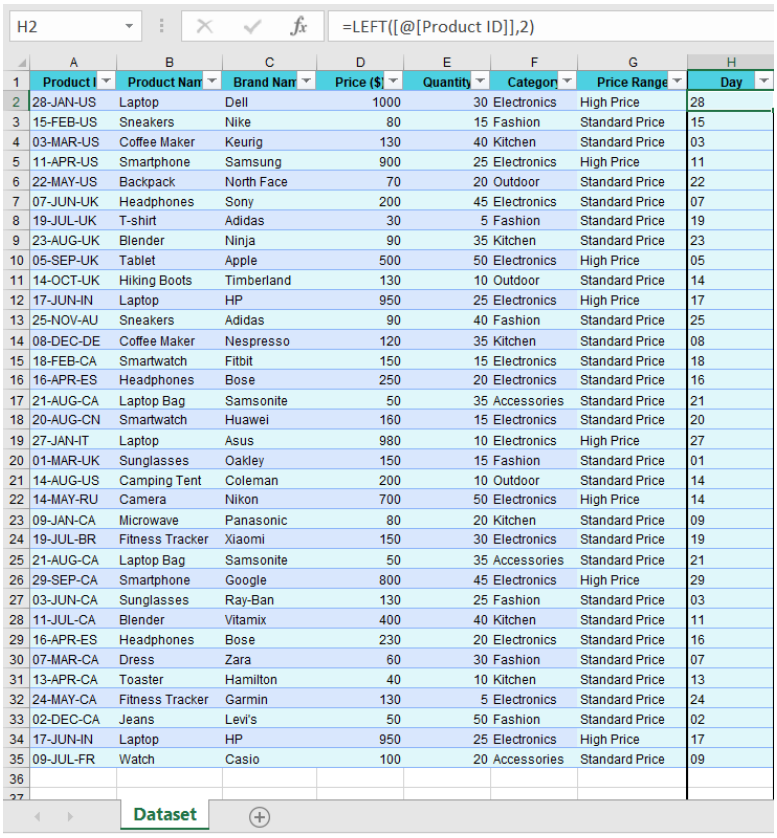
USING COUNTIF FORMULA



	F	G	H	I	J
16	Electronics	Standard Price	16	ES	APR
17	Accessories	Standard Price	21	CA	AUG
18	Electronics	Standard Price	20	CN	AUG
19	Electronics	High Price	27	IT	JAN
20	Fashion	Standard Price	01	UK	MAR
21	Outdoor	Standard Price	14	US	AUG
22	Electronics	High Price	14	RU	MAY
23	Kitchen	Standard Price	09	CA	JAN
24	Electronics	Standard Price	19	BR	JUL
25	Accessories	Standard Price	21	CA	AUG
26	Electronics	High Price	29	CA	SEP
27	Fashion	Standard Price	03	CA	JUN
28	Kitchen	Standard Price	11	CA	JUL
29	Electronics	Standard Price	16	ES	APR
30	Fashion	Standard Price	07	CA	MAR
31	Kitchen	Standard Price	13	CA	APR
32	Electronics	Standard Price	24	CA	MAY
33	Fashion	Standard Price	02	CA	DEC

5. Text Formatting – LEFT, RIGHT, MID

USING LEFT FORMULA

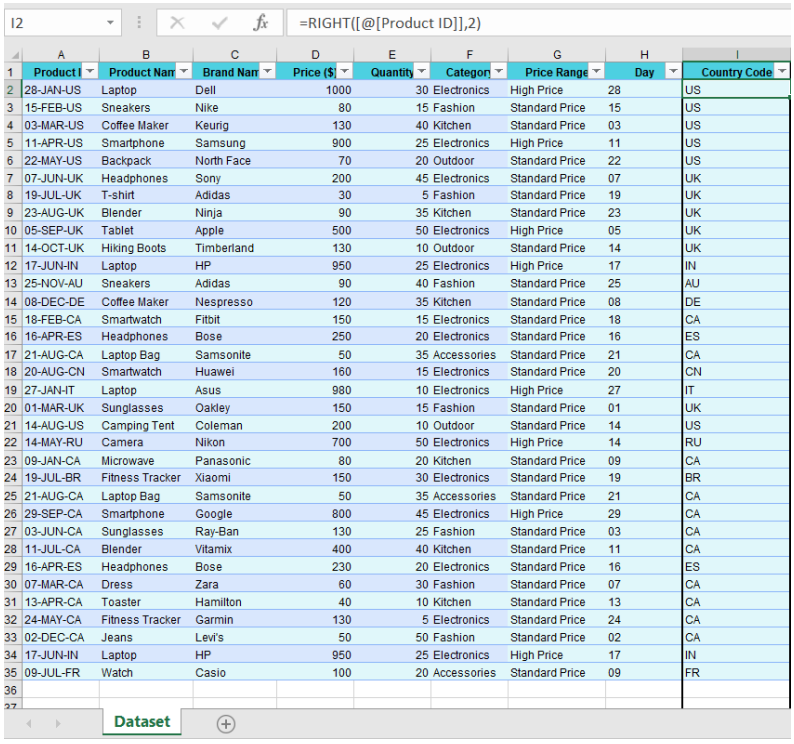


H2 =LEFT([@[Product ID]],2)

	A	B	C	D	E	F	G	H
1	Product I	Product Nam	Brand Nam	Price ($)	Quantity	Categor	Price Range	Day
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17
35	09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price	09
36								

Dataset

USING RIGHT FORMULA



I2 =RIGHT([@[Product ID]],2)

	A	B	C	D	E	F	G	H	I
1	Product I	Product Nam	Brand Nam	Price ($)	Quantity	Categor	Price Range	Day	Country Code
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN
35	09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price	09	FR
36									

Dataset

USING MID FORMULA

J2 =MID([@[Product ID]],4,3)

	A	B	C	D	E	F	G	H	I	J
1	Product I	Product Nan	Brand Nan	Price ($)	Quantity	Categor	Price Range	Day	Country Code	Month
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price	05	UK	SEP
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price	14	UK	OCT
12	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	Standard Price	25	AU	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	Standard Price	08	DE	DEC
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	Standard Price	18	CA	FEB
16	16-APR-ES	Headphones	Bose	250	20	Electronics	Standard Price	16	ES	APR
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	Standard Price	20	CN	AUG
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	High Price	27	IT	JAN
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	Standard Price	01	UK	MAR
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	Standard Price	14	US	AUG
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	High Price	14	RU	MAY
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	Standard Price	09	CA	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	Standard Price	19	BR	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	Standard Price	21	CA	AUG
26	29-SEP-CA	Smartphone	Google	800	45	Electronics	High Price	29	CA	SEP
27	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	Standard Price	03	CA	JUN
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	Standard Price	11	CA	JUL
29	16-APR-ES	Headphones	Bose	230	20	Electronics	Standard Price	16	ES	APR
30	07-MAR-CA	Dress	Zara	60	30	Fashion	Standard Price	07	CA	MAR
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	Standard Price	13	CA	APR
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	Standard Price	24	CA	MAY
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion	Standard Price	02	CA	DEC
34	17-JUN-IN	Laptop	HP	950	25	Electronics	High Price	17	IN	JUN
35	09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price	09	FR	JUL
36										

Dataset